



Department of Information Technology
Academic Year 2024 - 2025 (Even Semester)

Semester, Degree & Branch: VI Semester, B.Tech. IT

Course Code & Title: CCS339 & Cryptocurrency and Blockchain Technologies

Name of the Faculty member (s): Mrs. G.Sivasathiya, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit I / **Properties of Blockchain and Merkle Tree**
- **Course Outcome:** CO1
- **Topic Learning Outcome :** TLO 2 , TLO 5
- **Activity Chosen:** Mind Map
- **Justification:**

Blockchain concept has been explained completely with various features and technologies. A mind map illustrates the connections between the many parts of the total. It's usually made around a single idea, depicted as an image in the middle of a blank page, to which words, images, and word fragments are added as related illustrations of the notion. In order for students to comprehend the subject matter and organize information, they must be able to visualize the concepts.

- **Time Allotted for the Activity:** 15 Minutes

- **Details of the Implementation:**

- The time allotted for the topics is 15 Minutes and two topics have been chosen.
- The students were grouped into two bunches for drawing two mind maps.
- One group was given with the topic “**Properties of Blockchain**” and the other group was given with “**Merkle Tree**” topics and asked to draw mind maps.
- Based on their understanding with what they learned from the topics given was neatly presented in their papers in the format of Mind Maps.
- The students completed the activity, the sample of activity is shown in Figure 1 and 2, each for one topic.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO5	PO9	PO12	PSO1	PSO2	PSO3
CO1	3	3	2	2	3	2	2	1	1	3

(1 – Low

2 – Moderate

3 – High)

• **PO/PSO Mapped:**

Innovative Practice	PO1	PO2	PO3	PO4	PO5	PO9	PO12	PSO1	PSO2	PSO3
	3	3	2	2	3	2	2	1	1	3
Justification for Correlation	Students will acquire in-depth knowledge of the abstract models of blockchain technology by applying engineering fundamentals.	Students will analyze blockchain use cases and scenarios using engineering sciences.	Students will design and develop blockchain-based solutions by implementing smart contracts and decentralized applications.	Students will investigate complex problems related to abstract blockchain models, exploring potential improvements and innovative approaches	Students will effectively use blockchain development tools and platforms to implement and gain hands-on experience in blockchain technology	Students will collaborate effectively in teams to implement blockchain solutions, working together to address real-world challenges	Students will cultivate a mindset of continuous learning, staying updated with the latest advancements in blockchain technology to adapt to evolving industry trends and challenges.	To apply analytic technologies to arrive at actionable foresight, Insight, hindsight from data for solving business and engineering problems	To enrich the critical thinking skills in emerging technologies such as Hybrid Mobile application development, cloud technology stack, and cyber-physical systems with mathematical aid to foresee the research findings and provide the solutions.	To new trending strategies in software projects using open source environments and deliver a quality product for business and industry , banking sectors and for government organizations inorder to provide a secured and reliable transactions

- Images / Screenshot of the practice:

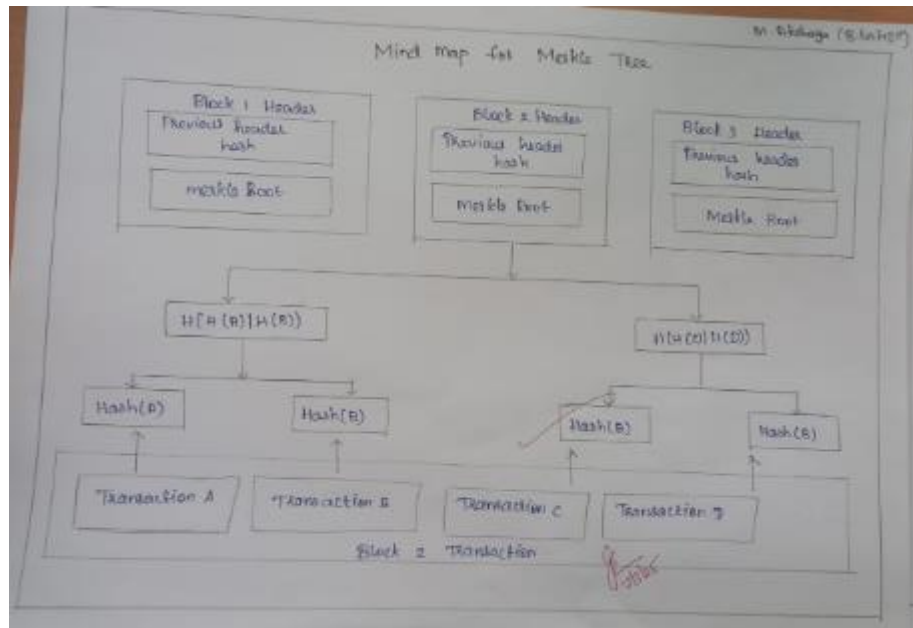


Figure 1. a Sample Mind - Map of Akshaya M (Merkle Tree)

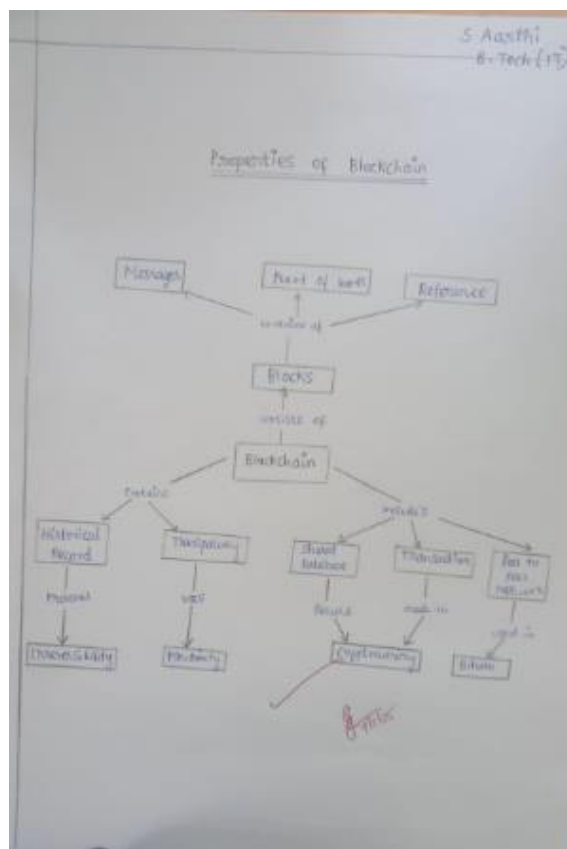


Figure 1. b Sample Mind-Map of Aarthi S (Properties of Blockchain)

Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

- The students felt that they can able to understand the basics of blockchain and how it works clearly after conducting this activity.
- The students reported that the activity helped them remember everything that was covered in the given topic and that it enhanced their ability to learn on their own.

❖ *Benefit of the practice:*

- Students were actively participated in this activity.
- Students can able to create their own views with the help of Mind Map.
- Make the students to know the impact and importance of understanding the topics in a diagrammatic / hierarchical representation related to the topic and made them involve in the activity.

❖ *Challenges faced in implementation:*

- Few students have drawn the map without any continuity and connectivity to the previous stage in the hierarchy.
- Few of them need to have some classification knowledge and clear exposing ability to draw the map neatly.
- After the final discussion, they acquired a clarity for the same.

References:

- <https://equip.eu/activity/activity-5-mind-mapping/>
- <https://nulab.com/learn/strategy-and-planning/mind-maps-guide/>
- <https://www.creativity-portal.com/articles/think-buzan/mind-mapping-for-innovation.html>

Signature of Faculty Member

HOD